FOAI — Assignment - 6

Q.1 Let:

- food(x) ⟶ x is food
- Eats(x, y) ⟶ x eats y
- Likes(x, y) → x likes y

Predicate Logic:

- $\forall x \forall y$ Eats(x, y) ⟶ food(y)
- $\forall y$ food(y) ⟶ likes(milka, y)
- food(bread)
- food(mangos)
- Eats(Alka, pizza)
- $\forall y$ Eats(Milka, y) ⟶ Eats(Alka, y)

CNF:

- ¬ Eats(x, y) ∨ food(y)
- ¬ food(y) ∨ Likes(Milka, y)
- ¬ Eats(Milka, y) ∨ Eats(Alka, y)

⇒ we need to prove likes(Milka, pizza)

Eats(Alka, pizza) ¬ Eats(Alka, pizza) ∨ food(pizza)

¬ food(pizza) ∨ Likes(Milka, y) food(pizza)

Likes(Milka, pizza).

⇒ Eats(Alka, pizza) ⟶ food(pizza)

so food Eaten by Alka is Pizza.

Q.2 Predicates :

- Child (x)
- Loves (x, y)
- Reindeer (x)
- Rednose (x)
- weird (x)
- Clown (x)

Constants :

- Santa
- Rudolph
- Scrooge.

Logic

- $\forall x$ Child $(x) \longrightarrow$ Loves (x, Santa)
  ↳ $\neg$ Child $(x) \vee$ Loves (x, Santa)
- $\forall x \forall y$ Loves $(x, \text{Santa}) \wedge$ Reindeer $(y) \rightarrow$ Loves (x, y)
  ↳ $\neg$ Loves $(x, \text{Santa}) \vee \neg$ Reindeer $(y) \vee$ Loves (x, y)
- Reindeer (Rudolph)
- Red Nose (Rudolph)
- $\forall x$ Red Nose $(x) \longrightarrow$ weird $(x) \vee$ Clown (x)
  ↳ $\neg$ Red Nose $(x) \vee$ weird $(x) \vee$ Clown (x)
- $\forall x$ Reindeer $(x) \longrightarrow \neg$ Clown (x)
  ↳ $\neg$ Reindeer $(x) \vee \neg$ Clown (x)
- $\forall x$ weird $(x) \longrightarrow \neg$ Loves (Scrooge, x)
  ↳ $\neg$ weird $(x) \vee \neg$ Loves (Scrooge, x)

we need to prove $\neg$ Child (Scrooge)

lets assume Child (Scrooge) is true.

$\neg$ Child (x) $\vee$ Loves (x, Santa)

Child (Scrooge)

$\neg$ Loves (x, Santa) $\vee$
$\neg$ Reindeer(y) $\vee$ Loves (x,y)

Loves (Scrooge, Santa)

$\therefore$ Loves (Scrooge, Rudolph)

Red Nose (Rudolph)

$\neg$ RedNose (x) $\vee$
weird (x) $\vee$ Clown (x)

weird (Rudolph) $\vee$
Clown (Rudolph)

Reindeer (Rudolph)

$\neg$ Reindeer (x) $\vee$
$\neg$ Clown (x)

weird (Rudolph) $\vee$ Clown (Rudolph)

$\neg$ Clown (Rudolph)

$\neg$ weird (x) $\vee$ $\neg$ Loves (Scrooge, x)

weird (Rudolph)

Loves (Scrooge, Rudolph)

$\neg$ Loves (Scrooge, Rudolph)

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So our contradiction was wrong.

$\therefore$ $\neg$ Child (Scrooge).